Islamic University of Technology

Machine Learning Lab

CSE 4622

Lab 2: Data Preprocessing and EDA

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1 Task 1: Data Loading and Inspection

The Titanic dataset was successfully loaded and inspected. The dataset contains both numerical and categorical features. Missing values were identified in Age, Cabin, and Embarked columns. Additionally, passengers who paid more than the average fare were identified.

2 Task 2: Handling Missing Values

Data preprocessing steps included:

- Dropping the Cabin column due to excessive missing values.
- Filling missing Age values with the median.
- Filling missing Embarked values with the mode.

After these operations, no missing values remained in the dataset.

3 Task 3: Encoding and Scaling

- Label Encoding was applied to the Sex column.
- One-Hot Encoding was applied to the Embarked column.
- Min-Max Scaling was applied to Age and Fare.
- Standardization (Z-score scaling) was also performed.

These steps converted all features into a numerical format suitable for machine learning.

4 Task 4: Exploratory Data Analysis

The dataset was explored using multiple visualizations to understand patterns and relationships among features.

- **Fare Distribution:** The histogram of Fare shows a right-skewed distribution, indicating that most passengers paid relatively low fares, while a small number paid significantly higher amounts.

- **Survival Count:** The bar chart of survival indicates that a larger proportion of passengers did not survive compared to those who survived.
- **Fare vs Passenger Class:** The box plot reveals that first-class passengers paid significantly higher fares than second- and third-class passengers, with noticeable outliers in the first class.
- **Correlation Heatmap:** The heatmap highlights relationships between numerical variables. Survival is strongly correlated with gender (Sex) and passenger class (Pclass), suggesting these factors influenced survival outcomes.
- **KDE Plot (Age by Gender):** The KDE plot comparing age distributions of male and female passengers shows that both groups have broadly similar distributions, with peaks around young adult ages. However, female passengers show slightly higher density in younger age ranges, while male passengers have a more spread-out distribution.

5 Task 5: Open-Ended Investigation

Question: Was the famous "women and children first" policy reflected in the Titanic survival data?

Approach

Passengers were grouped based on gender and age:

- Female Child (Age < 16)
- Female Adult (Age ≥ 16)
- Male Child (Age < 16)
- Male Adult (Age ≥ 16)

Visualization 1: Survival by Gender

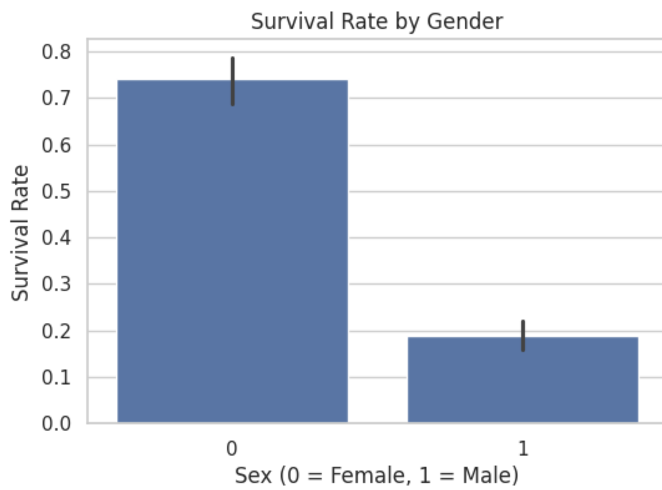

Figure 1: Survival Rate by Gender

Observation: Female passengers have a significantly higher survival rate than male passengers.

Visualization 2: Survival by Age Group

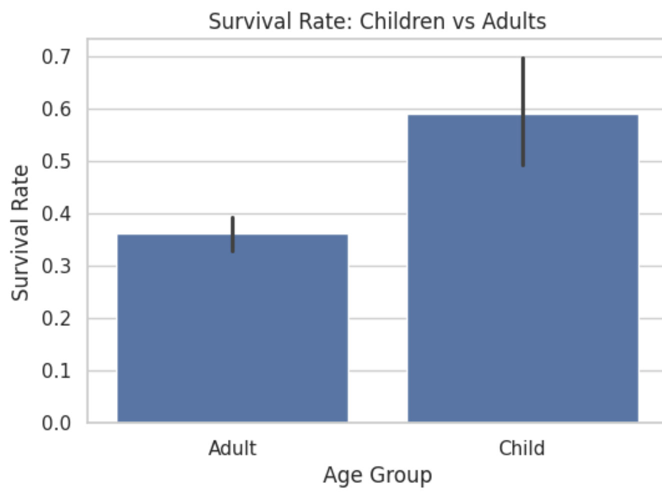

Figure 2: Survival Rate: Children vs Adults

Observation: Children have a higher survival rate compared to adults, indicating priority during evacuation.

Visualization 3: Survival by Gender and Age Group

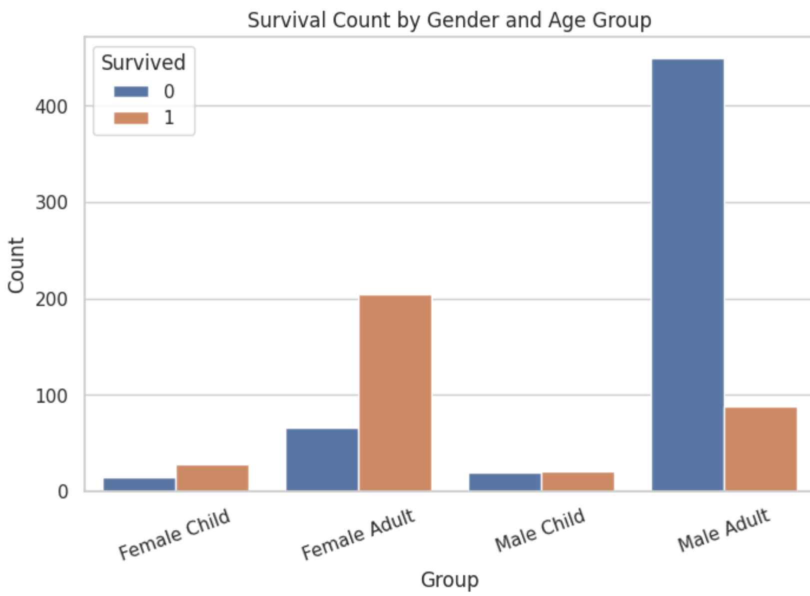

Figure 3: Survival Count by Gender and Age Group

Observation:

- Female children have the highest survival rate
- Female adults also have high survival rates
- Male children have moderate survival rates
- Male adults have the lowest survival rates

Conclusion

The analysis clearly supports the "women and children first" policy. Women had the highest survival rates, followed by children. Adult men were the least likely to survive.

All three visualizations consistently show that priority was given to women and children during evacuation, which aligns with historical accounts of the Titanic disaster.